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REQUEST FOR A THREE-YEAR ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

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Risk of external debt distress	Moderate[1]
Overall risk of debt distress	Moderate
Granularity in the risk rating	Limited space to absorb shocks
Application of judgment	No

The authorities have responded to the COVID-19 shock with fiscal support measures leading to a temporary widening of the fiscal deficit and an increase in public debt. Given the planned unwinding of crisis measures and the implementation of fiscal consolidation, Uganda's public debt continues to be sustainable in the medium term.

With debt-carrying capacity revised from strong to medium, Uganda has shifted from low to moderate risk of external and overall public debt distress, with limited space to absorb shocks, relative to the DSA prepared for the disbursement under the Rapid Credit Facility in May 2020. All external PPG debt and total public debt burden trajectories remain below, albeit closer than in the last DSA to their respective indicative thresholds and benchmarks over the medium term under the baseline scenario. Nevertheless, stress tests highlight breaches of external debt burden thresholds and the public debt benchmark, especially in relation to export shocks. Key risks include a slower-than-expected recovery from COVID-19, an increased frequency of climate disasters, higher current spending jeopardizing the consolidation in expenditures, incomplete implementation of the Domestic Revenue Mobilization Strategy (DRMS), slow execution of investment projects, further delays, beyond 2025, in oil exports coming on stream, a shift in the composition of financing towards non-concessional loans, and a potential decline in the capacity and appetite of commercial banks to provide financing.

[1] Uganda's Composite Indicator signals a medium debt-carrying capacity based on the October 2020 WEO and CPIA 2019.

PUBLIC DEBT COVERAGE

1. Public and publicly guaranteed (PPG) external and domestic debt covers debt contracted and guaranteed by the central government, state and local government, social security fund, and central bank (Text Table 1). The Uganda Public Debt Management Framework 2013 gives the Ministry of Finance, Planning and Economic Development the mandate to prepare and publish quarterly Debt Statistical Bulletins. The published data covers PPG debt with information on domestic and external debt. In addition, the Bank of Uganda (BoU) provides data on locally issued government debt held by non-residents, which allows a residency-based analysis. However, due to data limitations, debt data does not cover several elements of the general government debt including extra budgetary funds and non-guaranteed debt issued by state-owned enterprise (SOE), although SOEs issue debt only in the domestic market. For the contingent liability stress test, the framework includes the disputed arrears to Tanzania (US$58 million or 0.1 percent of GDP),[2] estimates of SOE debt (9.1 percent of GDP) based on a preliminary report by AFRITAC East, the default PPP shock (i.e., 35 percent of PPP stock, implying 1.7 percent of GDP), and the default financial market shock (5 percent of GDP).

Text Table 1. Uganda: Coverage of Public and Publicly Guaranteed Debt and Parameters for Contingent Liability Shock

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes
Subsectors of the public sector	
Central government	X
State and local government	X
Other elements in the general government	
o/w: Social security fund	X
o/w: Extra budgetary funds (EBFs)	
Guarantees (to other entities in the public and private sector, including to SOEs)	X
Central bank (borrowed on behalf of the government)	X
Non-guaranteed SOE debt	

The contingent liability tailored stress test

	Default	Used for the analysis
Other elements of the general government not captured in 1.	0 percent of GDP	0.1
SoE's debt (guaranteed and not guaranteed by the government)	2 percent of GDP	9.1
PPP	35 percent of PPP stock	1.7
Financial market (a minimum starting value of 5 percent of GDP)	5 percent of GDP	5
Total (in percent of GDP)		16.0

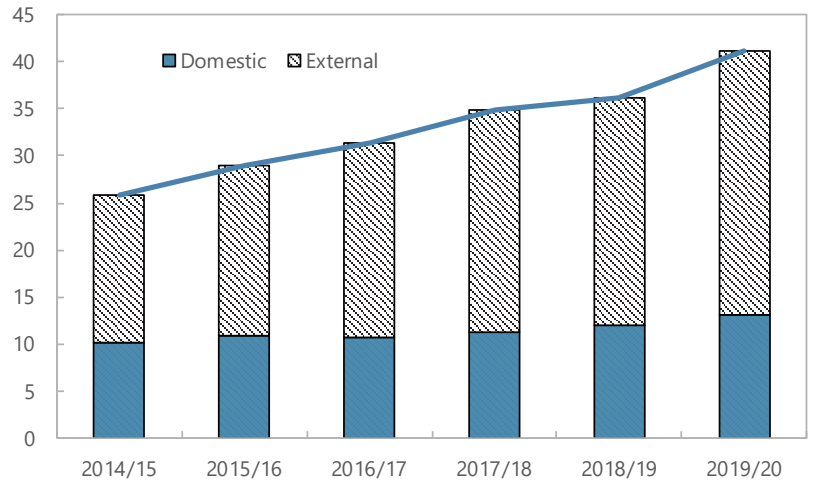
Source: Uganda authorities, IMF staff calculation.

[2] The arrears to Tanzania date back to the Uganda-Tanzania War in 1978-79, with an alleged lack of documentation of the debt, therefore the validity of these arrears is disputed and not included in total external debt. In contrast, arrears to Iraq (US$1,581) and Nigeria (US$19.3 million) are recorded in the official debt statistics. The Iraq and Nigeria arrears are pre-HIPC Initiative arrears to non-Paris Club creditors, which continue to be deemed away under the revised arrears policy for official creditors, as the underlying Paris Club agreement was adequately representative and the authorities continue to make best efforts to resolve the arrears.

BACKGROUND AND RECENT DEVELOPMENTS

2. Public debt has been increasing, reaching 40.8 percent of GDP at the end of FY2019/20 (Text Figure 1). The increase of almost six percentage points in FY2019/20 was primarily driven by external borrowing, with two-thirds of outstanding public debt owed to external creditors (US$10.7 billion or 28.6 percent of GDP) on a residency basis. Domestic debt amounts to about US$4.6 billion (12.3 percent of GDP). In present value terms, total public sector debt amounted to 33.2 percent of GDP at the end of FY2019/20.

Text Figure 1. Uganda: Public and Publicly Guaranteed Debt
(Percent of GDP)



Source: Uganda authorities; and IMF staff calculation.

3. While most of the existing stock of external public debt is on concessional terms, the semi-concessional component has been on the rise in recent years. Highly concessional loans from the IMF, the International Development Association (IDA) and the African Development Fund (ADF) account for half of the external debt portfolio, which drives the difference between the nominal value of public debt and its present value (Text Table 2). Other concessional creditors include the International Fund for Agricultural Development (IFAD), the Arab Bank for Economic Development in Africa (BADEA), the Organization of the Petroleum Exporting Countries (OPEC) fund, and some bilateral Paris and non-Paris club creditors. The share of semi-concessional and non-concessional debt has been increasing in recent years. The increase in semi-concessional loans has been driven by official loans from the Export-Import Bank of China, reaching 22 percent of external public debt outstanding at the end of FY2019/20. In response to COVID-19, Uganda also resorted to commercial loans that constitute 7 percent of external public debt, mostly owed to the Trade Development Bank, the Standard Bank of South Africa, and Standard Chartered. Finally, local-currency government securities held by offshore investors amounted to close to 3 percent of external public debt.

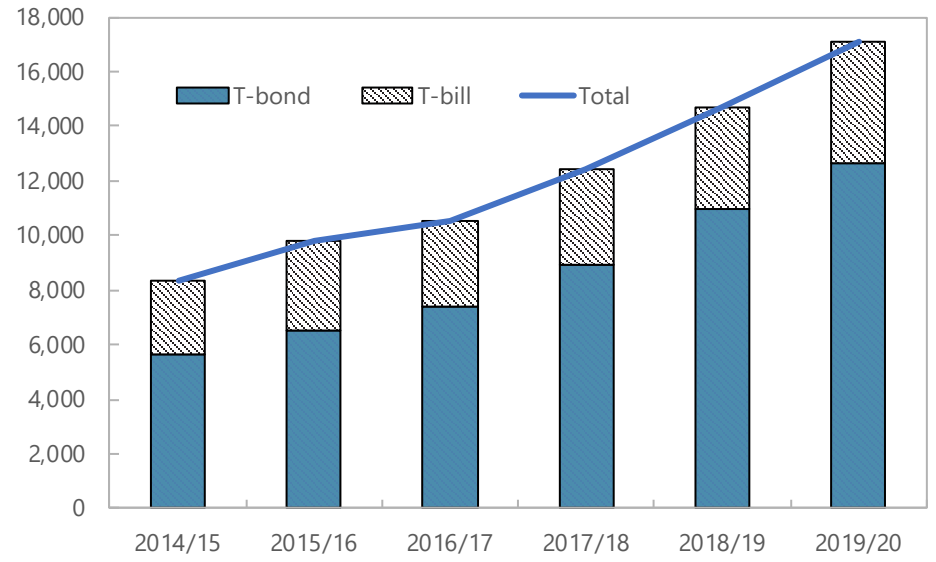
Text Table 2. Uganda: Composition of Public and Publicly Guaranteed External Debt
(Millions of U.S. dollars, as of end-FY2019/20)

Creditor	Amount	Share
Total	**10,702**	**100.0%**
Bilateral	3,203	29.9%
ABU DHABI FUND	8	0.1%
AFD	167	1.6%
EXIM BANK OF CHINA	2,362	22.1%
EXIM BANK S KOREA	24	0.2%
GOVT OF NIGERIA	12	0.1%
IRAQ FUND	0	0.0%
JBIC	88	0.8%
JICA	222	2.1%
KUWAIT FUND	36	0.3%
MIN FOR AFF AUSTRIA	3	0.0%
SAUDI ARABIA FUND	35	0.3%
UKEF	245	2.3%
Commercial banks or other financial institutions	748	7.0%
AKA	8	0.1%
COMMERZBANK	2	0.0%
SBSA	280	2.6%
STANDARD CHARTERED	121	1.1%
TDB (PTA)	336	3.1%
Multilateral	6,421	60.0%
ADB	140	1.3%
ADF	1,364	12.7%
BADEA	61	0.6%
EIB	105	1.0%
IDA	3,601	33.6%
IDB	296	2.8%
IFAD	248	2.3%
IMF	498	4.6%
NDF	50	0.5%
OPEC FUND	58	0.5%
Publicly guaranteed external debt	29	0.3%
Local currency debt held by offshore investors	302	2.8%

Source: Uganda authorities, IMF staff calculation.

4. Public domestic debt (residency based) is dominated by medium-to long-term securities. T-bonds constituted around three-fourths of public domestic debt at the end of FY2019/20 (Text Figure 2). Moreover, almost half of T-bonds have a maturity of at least 10 years. T-bonds are mostly held by non-banks (around 70 percent), while T-bills were predominantly held by banks (around 80 percent).

Text Figure 2. Uganda: Public Domestic Debt
(Cost value, billions of Ugandan Shillings)



Source: Uganda authorities; and IMF staff calculation.

MACRO FORECASTS

5. The medium-and long-term macroeconomic framework underlying this DSA is consistent with the scenario presented in the staff report (Text Table 3). The baseline scenario assumes the following:

- *Real GDP growth*. The growth outlook largely depends on the global recovery from the COVID-19 shock. In Uganda, 900,000 vaccines were received to cater for the most vulnerable groups (i.e., health workers, security personnel, and ages above 50 years). A stronger vaccination effort will take place in the second half of 2021 and well into 2022, with over US$120 million budgeted for purchases and administration of vaccines in FY21/22. Growth is expected at 3.3 percent in FY2020/21, half a percentage point below the projection in the RCF–less than half of the historical average–as external demand recovers only gradually. Over the medium term, growth is projected to return to above 6-7 percent, helped by an acceleration in private consumption due to pent-up demand, and the Final Investment Decision driving foreign and domestic private investment in oil infrastructure, in preparation of the start of oil production in FY2024/25.

- *Inflation*. Following an initial increase driven by demand recovery, inflation is projected to be in line with the BoU's 5 percent target over the medium term.

- *Oil revenue projections*. The Final Investment Decision was taken in April 2021 improving the likelihood of oil production in FY2024/25. Budget revenue net of oil-related expenditures is expected to peak at around 2 percent of GDP in FY2027/28 before gradually declining over the long term. Legislation and institutions have already been established, including the Uganda

Petroleum Fund, The Consolidated Fund, and the Petroleum Investment Fund to ensure fiscal sustainability over the longer run, which should help with managing the Dutch disease crowding out effects. The program also aims to help the authorities transition to a rules-based framework for oil revenue management.

- *Primary fiscal deficit.* The primary fiscal deficit is projected to widen in FY2020/21 partly on the back of COVID-19-related spending. In FY21/22, the large improvement is expected to be driven by the implementation of the DRMS,[3] a decline in both current and capital spending, including thanks to the unwinding of crisis measures, as well as the non-recurrence of one-off items such as the recapitalization of the Bank of Uganda and the on-lending to the Uganda Development Bank. After FY21/22, the cyclical improvement, the implementation of the DRMS, and the elimination of crisis measures will lead to a further improvement in the primary balance, notwithstanding a temporary deterioration associated with some large investment projects in FY24/25. Moreover, spending rationalization and greater spending efficiency under the ECF arrangement are expected to allow for higher social spending and the clearance of arrears during the program.

- *Debt Service Suspension Initiative.* External debt service of US$4 million and US$121 is assumed to be rescheduled from FY2020/21 to the period between FY2022/23 and FY2026/27, and from FY2021/22 to the period between FY2023/24 and FY2027/28, respectively, under the Debt Service Suspension Initiative.

- *Current account deficit.* The import-driven widening of the current account deficit in FY2020/21 will be offset by recovering tourism and remittance receipts from FY2021/22 onward, albeit subject to uncertainty around the pandemic and speed at which international tourism will rebound. This will be followed by a widening of the current account deficit on the back of oil-related investments until oil production expected in 2024/25 starts contributing to an improvement in the current account balance.

- *FDI inflows* are expected to continue to increase, more so than projected under the RCF, largely driven by investments in oil-related projects.

- *Gross official reserves* are expected to gradually rise over the medium term, on the back of an increase in FDI, the recovery of tourism, and the start of oil production

- *Financing mix.* Under the program, the limit on the present value of new external and publicly guaranteed external debt aims to both limit the accumulation of debt and prevent a heavy reliance on non-concessional external financing. Over the long term, financing is assumed to shift gradually towards less concessional debt.

[3] The implementation of the DRMS is projected to raise tax revenues by 0.5 percent of GDP per annum. Its approval is also a prior action under the IMF-supported arrangement.

Text Table 3. Uganda: Macroeconomic Assumptions

	Real GDP growth (percent change)		Primary balance (percent of GDP)		Current account (percent of GDP)		FDI inflows (percent of GDP)	
	RCF	Current	RCF	Current	RCF	Current	RCF	Current
2019/20	3.3	3.0	-5.0	-5.0	-10.1	-5.8	2.7	2.6
2020/21	3.7	3.3	-6.2	-7.1	-8.7	-9.2	2.9	2.0
2021/22	5.7	4.3	-3.0	-3.4	-5.5	-8.2	3.9	2.3
2022/23	6.0	6.4	-3.2	-0.9	-5.0	-8.6	4.1	5.3
2023/24	6.0	7.0	-2.5	-0.9	-3.8	-11.3	3.3	7.8
2024/25	9.2	7.2	-1.6	-1.3	-2.0	-13.2	3.6	9.7
2025/26	10.6	7.0	-1.2	-0.4	-0.5	-11.8	0.3	9.2
Avg 2026/27-2039/40	5.3	6.0	1.5	-0.9	-2.4	-7.0	0.2	4.0

Source: Uganda authorities, IMF staff projections.

6. The realism tool outputs compared the projections to cross-country experiences and to Uganda's own historical experience (Figures 3 and 4):

- There are small differences between past and projected debt creating flows. The contributions of past debt creating flows remain relatively the same for the projection period, and unexpected changes in public debt are near the median of the distribution across low-income countries. The smaller increase in average total public debt over the five-year projection horizon is accounted for by the unwinding of crisis measures and the fiscal consolidation under the ECF arrangement.

- Investment is expected to increase, with private investment offsetting a temporary decline in public investment. Relative to the last DSA, private investment has been revised upward, primarily due to higher estimated investments in the oil sector.

- The improvement in the primary balance over the next 3-years is in the top quartile of the distribution, reflecting the cyclical improvement in tax revenues, the unwinding of temporary crisis measures, as well as the adjustment following the fiscal policy response to COVID-19, including the implementation of the DRMS. Given that the first two factors are driven by the economic recovery from the COVID-19 shock, growth is expected to accelerate during the same period. Moreover, the growth outlook is also supported by the recovery in international demand and private investments as well as improved spending efficiency, including through stronger public investment management.

COUNTRY CLASSIFICATION

7. Uganda's debt-carrying capacity is classified as medium, a downgrade from strong. The classification of debt carrying capacity is guided by the composite indicator (CI) score, which is determined by the World Bank's CPIA and other variables, such as real GDP growth, import coverage of foreign exchange reserves, remittances, and growth of the world economy. The CI also incorporates forward-

looking elements with the calculation based on the 10-year average (5 years of historical data and 5 years of projection). Uganda's latest CI is 2.95, which lies between the threshold values of 2.69 and 3.05 corresponding to medium and strong capacity, respectively, thereby categorizing the country as having "medium" debt-carrying capacity. The resulting downgrade from "strong" at the time of the 2019 Article IV and the May 2020 RCF is mostly driven by lower global growth, followed by lower reserves and reduced growth in Uganda. As a result, the four external indicative thresholds and the total public debt benchmark, determined by the classification of the debt carrying capacity, have been adjusted downward (Text Table 4).

Text Table 4. Uganda: Composite Indicator, and Debt Thresholds and Benchmark

Debt Carrying Capacity	Medium

Final	Classification based on Oct 2020 vintage	Classification based on Oct 2019 vintage
Medium	Medium 2.95	Strong 3.08

EXTERNAL debt burden thresholds	Weak	Medium	Strong
PV of debt in % of			
Exports	140	180	240
GDP	30	40	55
Debt service in % of			
Exports	10	15	21
Revenue	14	18	23
TOTAL public debt benchmark	**Weak**	**Medium**	**Strong**
PV of total public debt in percent of GDP	35	55	70

Source: IMF staff projections.

EXTERNAL DEBT SUSTAINABILITY

8. The evolution of external government debt suggests a sustainable path under the baseline (Table 1). Both solvency and liquidity indicators remain below their indicative thresholds over the projection horizon (Figure 1 and Table 3). Specifically, the PV of PPG external debt-to-GDP ratio peaks at 27.5 percent (against the threshold of 40 percent), while the PV of debt-to-exports ratio reaches 160.8 percent before gradually declining below 150 percent over the medium term (against the threshold of 180 percent). The liquidity indicators show a similar path. The debt service-to-revenue ratio peaks at 15.4 percent, well below the threshold of 18 percent, while the debt service-to-exports ratio is close to a one-off breach with its peak reaching 14.9 percent in FY29/30 against the indicative threshold of 15 percent. The weak export base is a source of potential debt vulnerabilities as both debt burden trajectories (debt-to-exports and debt service-to-exports) are on an upward trend after the first two years of oil production.

9. Stress tests and alternative scenarios indicate a moderate risk of debt distress rating. The PV of PPG debt-to-GDP ratio and the debt-service-to-revenue ratio remain under their respective thresholds even under the stress tests (apart from a short-lived breach in 2022/23 in the case of the latter), with the combined contingent liability shock and the depreciation being the most extreme shock, respectively. At the same time, the PV of debt-to-exports ratio and the debt service-to-exports ratio breach their respective

thresholds under several stress tests (Figure 1). Specifically, for the PV of debt-to-exports ratio and debt service-to-exports ratio, the most extreme shock is a shock to exports, with long-lasting breaches of the thresholds. Both indicators breach their thresholds under the combined shock and the combined contingent liability shock.

PUBLIC DEBT SUSTAINABILITY

10. The total public debt-to-GDP trajectory under the baseline shows a declining path (Table 2). The PV of public debt-to-GDP ratio peaks at 43.7 percent in FY2021/22, before declining towards 30 percent by FY2030/31 as oil export receipts ensue (Figure 2 and Table 4). This compares to an indicative benchmark of 55 percent for countries with medium debt-carrying capacity.[4] Notwithstanding the increase in the nominal level of public debt-to-GDP to above 50 percent of GDP over the next few years, the trajectory is expected to decline gradually over the medium-and long-run. The PV of debt-to-revenue ratio and the debt service-to-revenue ratio are expected to decline over the medium term, supported by the implementation of the DRMS and the oil-related revenue inflows from 2024/25 onward. Putting these indicators on a declining path is especially important given that commercial banks' capacity to purchase government securities could be negatively impacted by the increasing weight of those securities in their balance sheets.

11. The stress tests confirm that adding domestic debt to the analysis does not elevate the risk of debt distress. For the PV of debt-to-GDP ratio, the most extreme stress test is the contingent liability shock (Figure 2). Even under this shock scenario, however, the PV of total public debt-to-GDP ratio rises only slightly and temporarily above the benchmark of 55 percent and the ceiling of 50 percent of GDP in the Charter of Fiscal Responsibility. The debt burden indicator remains below the benchmark under all other shocks. The PV of total public debt-to-revenue ratio rises to slightly below 390 percent under the most extreme scenario of the contingent liabilities shock. The most extreme shock to public debt service is also the combined contingent liability shock, under which the debt service-to-revenue peaks at 70 percent.

CONCLUSIONS

12. Uganda's risk of external and overall public debt distress is moderate, with limited space to absorb shocks. External debt burden indicators and total public debt remain below their respective thresholds and benchmark throughout the projection horizon under the program scenario. The stress tests, however, indicate breaches of the thresholds and the benchmark. Although the PV of debt-to-GDP ratio and the debt service-to-revenue ratios indicate some space to absorb shocks without being downgraded to a high risk of debt distress, the PV of debt-to-exports and the debt service-to-exports ratios are close enough to its respective threshold that a median shock would lead to a breach (Figure 5).

[4] The government's Charter of Fiscal Responsibility requires public debt to stay below 50 percent of GDP in present value terms, which is also one of the convergence criteria for monetary union in the East African Community. The government's publicly stated debt ceiling is 50 percent of GDP in nominal terms.

13. Risks to the debt outlook are tilted to the downside. Given the sensitivity of the debt projections to growth and exports, a resurgence of domestic and external lockdown measures linked to another wave of COVID-19 constitutes a downside risk to the outlook. Risks around growth are further aggravated by the uncertainty around the execution of public investment projects, any further delay in oil production, and increased frequency of natural disasters due to climate change. Domestic risks include political pressures for higher current spending, poor prioritization of spending, the insufficient implementation of the DRMS, and a shift in the composition of financing towards non-concessional loans. On the upside, a faster than expected recovery from the pandemic and higher donor financing would help.

14. Mitigating debt risks requires sound macroeconomic management and strong/steadfast policy implementation. This includes:

- *Implementing the DRMS*. Given Uganda's relatively low revenue collection, the strategy is expected to include tax policy reforms, including a rationalization of exemptions, and tax administration reforms to improve compliance.

- *Strengthening overall public financial management (PFM), including efforts to avoid arrears*. The ongoing finalization of the international audit of domestic arrears (structural benchmark) should help the authorities in defining a clear time-bound repayment plan of past arrears by FY23/24. Complementary cash management reforms should include extending the Treasury Single Account to extra-budgetary units and externally funded projects and updating cashflow forecasts monthly as per international good practice.

- *Improving spending efficiency, including the strengthening of public investment management*. Priorities should include improving the use of medium-term fiscal envelope forecasts to achieve better project prioritization and capital expenditure budgeting. Publishing and adhering to clearly defined selection criteria is essential to avoid adding unplanned projects to the budget, and exercise rigorous public investment portfolio oversight.

- *Strengthening debt management*. In line with its medium-term debt strategy, public debt management in Uganda should continue to ensure that the government's financing needs, and its payment obligations are met at the lowest possible cost over the medium to long run, consistent with a prudent risk-taking. Better communication and coordination across government agencies, including on new borrowing plans, would further enhance debt management.

- *Improving debt transparency*. Over the last decade, Uganda has enhanced debt transparency, both in terms of fiscal reporting and publication of explicit and implicit debt and debt management information. However, debt transparency could be further enhanced through more data on state-owned enterprise debt and better communication.

- *Closely monitoring contingent liabilities*. Contingent liabilities have in general been one of the largest sources of fiscal risk across countries, since the materialization of contingent liabilities can contribute to unexpected increases in the debt-to-GDP ratio, crowding out private credit and jeopardizing debt sustainability. Efforts should be stepped up to estimate, disclose, manage, and

contain contingent liabilities, especially those in the financial sector, state-owned enterprises (including through their potential inclusion in government finance statistics), and PPPs.

- *Enhancing governance frameworks.* These are equally essential to safeguard the quality and effectiveness of public investment and other government spending. A better infrastructure and the impact of parallel reforms, e.g., improvements in the business climate, are expected to strengthen Uganda's competitiveness. Sound asset-liability management and avoidance of a premature reliance on uncertain future oil-related flows remain essential preconditions for debt sustainability.

AUTHORITIES' VIEWS

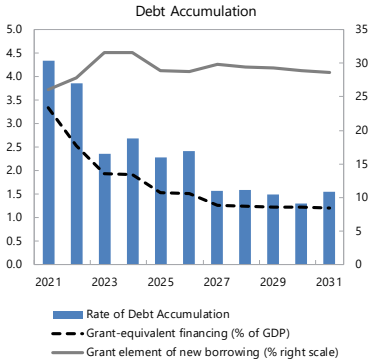
15. The authorities agreed with the assessment of the risk of debt distress. They also emphasized the importance of reducing public debt to below 50 percent of GDP over the medium term through the combination of the implementation of the DRMS and the unwinding of crisis measures introduced in response to COVID-19.

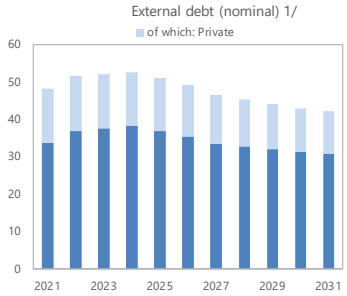
Table 1. Uganda: External Debt Sustainability Framework, Baseline Scenario, FY2017/18-2030/31

(In percent of GDP, unless otherwise indicated)

	Actual			Projections							Average 8/	
	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2030/31	Historical	Projections
External debt (nominal) 1/	**38.3**	**38.4**	**42.9**	**48.2**	**51.5**	**52.0**	**52.7**	**51.1**	**49.2**	**42.2**	**33.9**	**47.8**
of which: public and publicly guaranteed (PPG)	**24.4**	**24.1**	**28.6**	**33.7**	**36.9**	**37.5**	**38.2**	**36.8**	**35.4**	**30.7**	**19.9**	**34.8**
Change in external debt	2.6	0.1	4.5	5.3	3.3	0.5	0.7	-1.5	-1.9	-0.8		
Identified net debt-creating flows	**0.1**	**0.6**	**1.1**	**5.8**	**3.9**	**0.3**	**0.2**	**0.1**	**-0.5**	**1.6**	**0.0**	**1.3**
Non-interest current account deficit	**4.5**	**6.0**	**5.1**	**8.3**	**7.3**	**7.6**	**10.4**	**12.3**	**9.2**	**5.9**	**5.9**	**7.8**
Deficit in balance of goods and services	0.2	1.2	1.5	3.9	2.8	3.5	6.1	7.7	4.9	2.2	**1.1**	**3.6**
Exports	16.9	18.1	14.8	15.4	17.2	17.2	17.3	16.8	16.9	14.7		
Imports	17.1	19.3	16.4	19.3	20.0	20.7	23.5	24.4	21.8	16.8		
Net current transfers (negative = inflow)	-4.8	-4.9	-4.8	-4.3	-4.1	-4.7	-5.1	-4.9	-4.7	0.0	**-5.4**	**-4.5**
of which: official	-0.4	-0.4	-0.4	-0.8	-0.5	-0.3	-0.3	-0.2	-0.2	0.0		
Other current account flows (negative = net inflow)	9.2	9.8	8.4	8.6	8.6	8.8	9.4	9.6	8.9	3.8	**10.2**	**8.6**
Net FDI (negative = inflow)	**-2.8**	**-3.4**	**-2.6**	**-2.0**	**-2.3**	**-5.3**	**-7.8**	**-9.7**	**-9.2**	**-4.0**	**-3.4**	**-5.5**
Endogenous debt dynamics 2/	**-1.7**	**-2.0**	**-1.5**	**-0.4**	**-1.0**	**-2.1**	**-2.4**	**-2.5**	**-0.5**	**-0.3**		
Contribution from nominal interest rate	0.7	0.7	0.7	0.9	0.9	1.0	1.0	0.9	2.7	2.0		
Contribution from real GDP growth	-2.1	-2.3	-1.1	-1.3	-2.0	-3.1	-3.4	-3.4	-3.2	-2.3		
Contribution from price and exchange rate changes	-0.3	-0.4	-1.1	...	...	...	...	...	...	...		
Residual 3/	**2.5**	**-0.5**	**3.5**	**-0.5**	**-0.6**	**0.2**	**0.5**	**-1.6**	**-1.4**	**-2.4**	**0.8**	**-1.3**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators												
PV of PPG external debt-to-GDP ratio	**...**	**...**	**20.9**	**24.3**	**26.7**	**27.0**	**27.5**	**27.0**	**26.1**	**22.8**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**140.6**	**157.7**	**155.1**	**157.2**	**158.4**	**160.8**	**154.6**	**155.8**		
PPG debt service-to-exports ratio	**14.6**	**16.5**	**8.2**	**12.3**	**10.5**	**12.9**	**12.3**	**13.3**	**13.5**	**13.9**		
PPG debt service-to-revenue ratio	**20.5**	**23.7**	**9.8**	**14.4**	**13.0**	**15.4**	**14.3**	**14.3**	**13.3**	**11.7**		
Gross external financing need (Million of U.S. dollars)	1,769	2,523	2,818	4,785	4,510	3,686	4,159	4,380	4,479	7,309		
Key macroeconomic assumptions												
Real GDP growth (in percent)	6.3	6.4	3.0	3.3	4.3	6.4	7.0	7.2	7.0	5.8	**4.1**	**5.9**
GDP deflator in US dollar terms (change in percent)	0.7	1.1	3.1	0.6	1.1	1.0	1.1	2.8	5.1	2.0	**8.2**	**2.0**
Effective interest rate (percent) 4/	2.1	2.1	2.0	2.3	2.1	2.1	2.0	2.0	5.8	4.9	**1.8**	**2.7**
Growth of exports of G&S (US dollar terms, in percent)	12.0	15.5	-13.0	7.7	17.8	7.4	9.1	6.6	13.4	2.1	**5.3**	**10.3**
Growth of imports of G&S (US dollar terms, in percent)	17.9	21.5	-9.7	22.6	8.8	11.5	22.6	14.8	0.5	6.4	**4.6**	**13.5**
Grant element of new public sector borrowing (in percent)	...	...	...	26.1	27.8	31.6	31.6	28.9	28.7	28.7	**...**	**29.1**
Government revenues (excluding grants, in percent of GDP)	12.0	12.6	12.4	13.1	13.8	14.4	14.9	15.5	17.1	17.5	**12.2**	**14.8**
Aid flows (in Million of US dollars) 5/	-1,684	-1,576	-1,984	1,646	1,426	1,072	970	707	752	981		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	3.3	2.5	1.9	1.9	1.5	1.5	1.2	**...**	**2.1**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	39.1	37.4	38.5	36.6	34.2	32.6	33.5	**...**	**36.4**
Nominal GDP (Million of US dollars)	32,912	35,432	37,600	39,052	41,197	44,266	47,887	52,780	59,358	91,715		
Nominal dollar GDP growth	7.1	7.7	6.1	3.9	5.5	7.4	8.2	10.2	12.5	7.9	**12.0**	**7.9**
Memorandum items:												
PV of external debt 7/	...	...	35.2	38.8	41.2	41.5	41.9	41.3	40.0	34.3		
In percent of exports	...	...	237.3	251.8	239.7	241.4	241.8	246.1	236.3	234.1		
Total external debt service-to-exports ratio	21.2	25.1	33.3	39.3	35.0	34.9	35.0	33.8	44.8	41.2		
PV of PPG external debt (in Million of US dollars)			7,850	9,482	10,987	11,959	13,149	14,240	15,515	20,928		
(PVt-PVt-1)/GDPt-1 (in percent)				4.3	3.9	2.4	2.7	2.3	2.4	1.6		
Non-interest current account deficit that stabilizes debt ratio	1.9	5.9	0.6	3.0	4.0	7.1	9.7	13.8	11.1	6.8		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

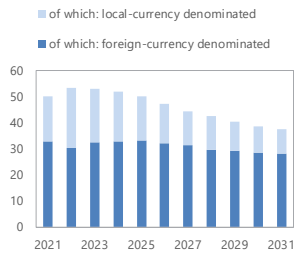
Table 2. Uganda: Public Sector Debt Sustainability Framework, Baseline Scenario, FY2017/18-2030/31

(In percent of GDP, unless otherwise indicated)

	Actual			Projections							Average 6/	
	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2030/31	Historical	Projections
Public sector debt 1/	**34.8**	**35.2**	**40.8**	**50.2**	**53.5**	**53.0**	**52.0**	**50.2**	**47.4**	**37.7**	**29.3**	**47.2**
of which: external debt	**24.4**	**24.1**	**28.6**	**33.7**	**36.9**	**37.5**	**38.2**	**36.8**	**35.4**	**30.7**	**19.9**	**34.8**
Change in public sector debt	3.5	0.4	5.7	9.4	3.3	-0.5	-1.0	-1.7	-2.9	-0.9		
Identified debt-creating flows	**2.9**	**0.8**	**5.2**	**9.5**	**3.3**	**-0.1**	**-0.7**	**-1.2**	**-2.3**	**-0.7**	**0.4**	**0.1**
Primary deficit	**2.2**	**3.0**	**5.0**	**7.1**	**3.4**	**0.9**	**0.9**	**1.3**	**0.4**	**0.7**	**2.1**	**1.3**
Revenue and grants	12.7	13.5	13.2	14.6	14.7	14.9	15.3	15.9	17.4	17.7	**13.4**	**16.5**
of which: grants	0.6	0.9	0.8	1.5	0.9	0.5	0.4	0.3	0.3	0.2		
Primary (noninterest) expenditure	14.9	16.5	18.2	21.7	18.1	15.8	16.2	17.2	17.8	18.4	**15.5**	**17.9**
Automatic debt dynamics	**0.6**	**-2.1**	**0.2**	**1.5**	**0.0**	**-1.0**	**-1.6**	**-2.5**	**-2.7**	**-1.4**		
Contribution from interest rate/growth differential	-0.5	-0.9	0.3	0.9	-0.2	-1.2	-1.8	-1.8	-1.3	-1.1		
of which: contribution from average real interest rate	1.4	1.2	1.3	2.1	1.8	2.0	1.7	1.7	1.9	1.0		
of which: contribution from real GDP growth	-1.9	-2.1	-1.0	-1.3	-2.1	-3.2	-3.5	-3.5	-3.3	-2.1		
Contribution from real exchange rate depreciation	1.1	-1.3	-0.1	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.1**	**0.0**	**0.0**	**1.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.1**	**0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**0.6**	**-0.4**	**0.4**	**-0.2**	**0.0**	**-0.4**	**-0.3**	**-0.6**	**-0.6**	**-0.2**	**0.8**	**-0.5**
Sustainability indicators												
PV of public debt-to-GDP ratio 2/	**...**	**...**	**33.2**	**41.2**	**43.7**	**43.0**	**41.7**	**40.4**	**38.0**	**30.0**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**251.6**	**282.0**	**296.8**	**288.8**	**272.6**	**254.7**	**219.1**	**169.4**		
Debt service-to-revenue and grants ratio 3/	**59.3**	**49.0**	**46.1**	**55.7**	**55.9**	**51.4**	**40.5**	**39.3**	**36.5**	**24.1**		
Gross financing need 4/	7.8	7.7	11.1	16.2	11.6	8.6	7.1	7.5	6.8	5.0		
Key macroeconomic and fiscal assumptions												
Real GDP growth (in percent)	6.3	6.4	3.0	3.3	4.3	6.4	7.0	7.2	7.0	5.8	**4.1**	**6.0**
Average nominal interest rate on external debt (in percent)	2.0	1.9	1.8	2.8	2.7	2.5	2.5	2.3	2.1	1.7	**1.5**	**2.2**
Average real interest rate on domestic debt (in percent)	-0.1	-0.2	0.3	1.3	0.8	0.6	0.6	0.3	1.1	0.7	**-0.2**	**0.8**
Real exchange rate depreciation (in percent, + indicates depreciation)	5.7	-5.6	-0.2	...	...	...	...	...	...	...	**-4.2**	**...**
Inflation rate (GDP deflator, in percent)	4.4	3.0	2.7	2.8	4.8	4.6	4.6	4.6	4.9	4.3	**14.6**	**4.3**
Growth of real primary spending (deflated by GDP deflator, in percent)	36.2	17.7	14.0	23.0	-13.0	-7.1	9.5	13.8	10.8	9.3	**3.4**	**6.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-1.3	2.6	-0.6	-2.3	0.0	1.4	1.9	3.0	3.3	1.6	**0.2**	**1.5**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus social security, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Uganda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, FY2020/21-2030/31

(In percent)

	Projections										
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	2027/28	2028/29	2029/30	2030/31
PV of debt-to GDP ratio											
Baseline	**24.3**	**26.7**	**27.0**	**27.5**	**27.0**	**26.1**	**24.8**	**24.0**	**23.6**	**23.1**	**22.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	24.3	23.8	23.4	23.2	22.5	23.3	22.3	22.4	22.3	21.6	20.8
B. Bound Tests											
B1. Real GDP growth	24.3	28.2	31.2	31.7	31.1	30.2	28.6	27.7	27.2	26.6	26.3
B2. Primary balance	24.3	27.4	28.6	29.1	28.6	27.7	26.4	25.7	25.3	24.7	24.4
B3. Exports	24.3	28.7	31.9	32.2	31.5	30.3	28.6	27.7	27.1	26.3	25.8
B4. Other flows 2/	24.3	28.5	30.5	30.8	30.2	29.1	27.5	26.6	26.1	25.3	24.9
B6. One-time 30 percent nominal depreciation	24.3	33.5	32.1	32.7	32.2	31.3	29.7	28.8	28.4	27.8	27.6
B6. Combination of B1-B5	24.3	31.1	34.1	34.4	33.7	32.5	30.8	29.7	29.1	28.3	27.9
C. Tailored Tests											
C1. Combined contingent liabilities	24.3	33.8	35.1	35.6	34.9	33.8	33.2	32.5	32.0	31.3	30.9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**40**	**40**	**40**	**40**	**40**	**40**	**40**	**40**	**40**	**40**	**40**
PV of debt-to-exports ratio											
Baseline	**157.7**	**155.1**	**157.2**	**158.4**	**160.8**	**154.6**	**139.7**	**142.6**	**145.8**	**149.0**	**155.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	157.7	138.7	136.2	133.6	134.0	137.8	125.8	133.2	137.5	139.3	141.7
B. Bound Tests											
B1. Real GDP growth	157.7	155.1	157.2	158.4	160.8	154.6	139.7	142.6	145.8	149.0	155.8
B2. Primary balance	157.7	159.4	166.3	168.0	170.6	163.9	148.9	152.8	156.2	159.5	166.4
B3. Exports	157.7	**206.0**	**258.2**	**258.1**	**260.6**	**248.9**	**224.3**	**228.4**	**232.5**	**235.9**	**244.7**
B4. Other flows 2/	157.7	165.5	177.7	177.9	179.8	172.0	155.1	158.0	160.9	163.5	169.9
B6. One-time 30 percent nominal depreciation	157.7	154.5	147.9	149.6	152.3	146.7	132.7	135.7	138.9	142.4	149.3
B6. Combination of B1-B5	157.7	**199.4**	174.6	**220.8**	**223.4**	**213.8**	**192.8**	**196.5**	**200.0**	**203.5**	**211.6**
C. Tailored Tests											
C1. Combined contingent liabilities	157.7	**196.7**	**204.1**	**205.3**	**208.2**	**199.6**	**187.2**	**192.8**	**197.5**	**202.1**	**210.9**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**
Debt service-to-exports ratio											
Baseline	**12.3**	**10.5**	**12.9**	**12.3**	**13.3**	**13.5**	**12.5**	**12.7**	**13.2**	**14.9**	**13.9**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	12.3	9.8	11.4	10.5	11.1	11.3	10.6	10.6	10.6	11.7	10.6
B. Bound Tests											
B1. Real GDP growth	12.3	10.5	12.9	12.3	13.3	13.5	12.5	12.7	13.2	14.9	13.9
B2. Primary balance	12.3	10.5	13.0	12.6	13.5	13.7	12.7	13.2	14.0	**15.8**	14.8
B3. Exports	12.3	13.0	**18.5**	**18.2**	**19.5**	**19.7**	**18.2**	**18.5**	**19.7**	**22.8**	**21.4**
B4. Other flows 2/	12.3	10.5	13.1	12.8	13.7	13.9	12.9	13.0	13.9	**15.9**	14.9
B6. One-time 30 percent nominal depreciation	12.3	10.5	12.9	12.1	13.1	13.3	12.3	12.5	13.1	14.5	13.5
B6. Combination of B1-B5	12.3	12.3	**16.7**	**16.1**	**17.3**	**17.5**	**16.2**	**16.4**	**17.6**	**19.9**	**18.6**
C. Tailored Tests											
C1. Combined contingent liabilities	12.3	10.5	14.0	13.6	14.5	14.7	13.5	13.9	14.5	**16.3**	**15.3**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**
Debt service-to-revenue ratio											
Baseline	**14.4**	**13.0**	**15.4**	**14.3**	**14.3**	**13.3**	**12.3**	**11.8**	**11.9**	**13.0**	**11.7**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	14.4	12.2	13.6	12.2	12.0	11.2	10.5	9.9	9.6	10.2	8.9
B. Bound Tests											
B1. Real GDP growth	14.4	13.8	17.8	16.5	16.5	15.4	14.2	13.6	13.8	15.1	13.5
B2. Primary balance	14.4	13.0	15.5	14.6	14.6	13.6	12.5	12.3	12.6	13.8	12.4
B3. Exports	14.4	13.1	15.9	15.2	15.2	14.0	12.9	12.4	12.8	14.3	12.9
B4. Other flows 2/	14.4	13.0	15.7	14.9	14.8	13.8	12.7	12.2	12.6	13.9	12.5
B6. One-time 30 percent nominal depreciation	14.4	16.4	**19.4**	17.7	17.8	16.6	15.3	14.7	14.9	16.0	14.3
B6. Combination of B1-B5	14.4	13.9	18.0	16.8	16.8	15.5	14.3	13.7	14.3	15.6	14.1
C. Tailored Tests											
C1. Combined contingent liabilities	14.4	13.0	16.8	15.8	15.7	14.5	13.3	13.0	13.1	14.2	12.8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**18**	**18**	**18**	**18**	**18**	**18**	**18**	**18**	**18**	**18**	**18**

Sources: Country authorities; and staff estimates and projections.

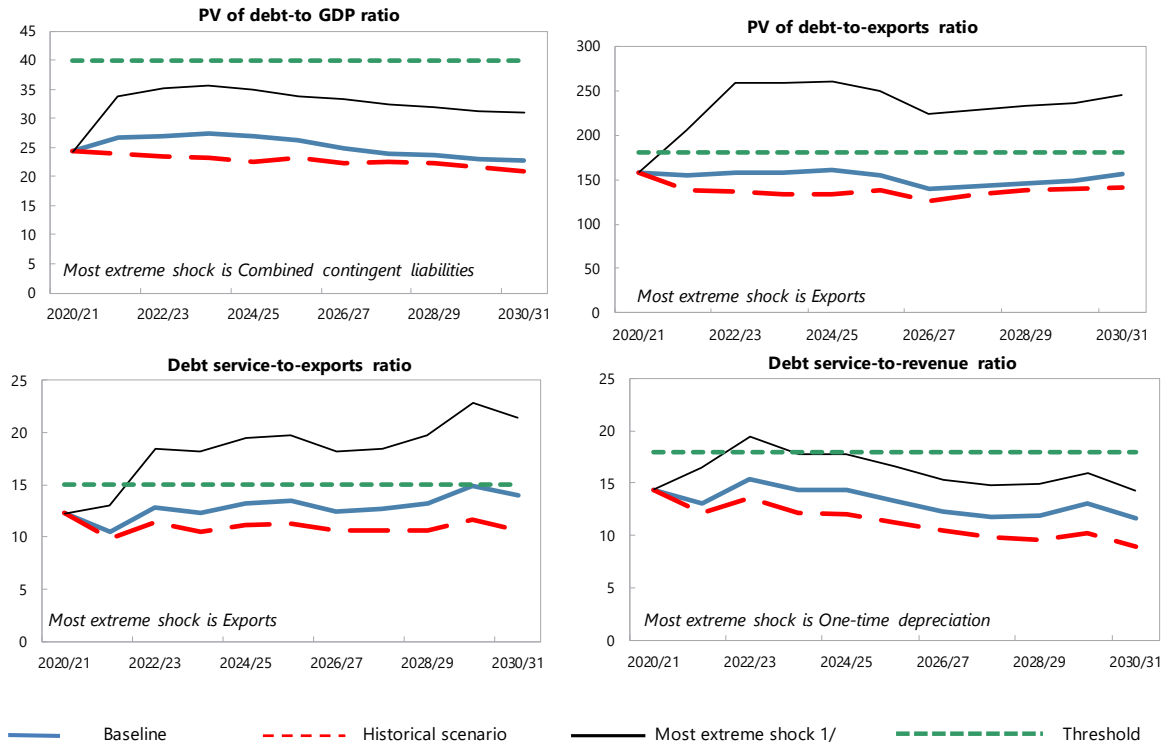
1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Includes official and private transfers and FDI.

Table 4. Uganda: Sensitivity Analysis for Key Indicators of Public Debt, FY2020/21-2030/31

	Projections										
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	2027/28	2028/29	2029/30	2030/31
PV of Debt-to-GDP Ratio											
Baseline	**41.2**	**43.7**	**43.0**	**41.7**	**40.4**	**38.0**	**35.8**	**34.1**	**32.3**	**30.7**	**30.0**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	41	39	37	34	32	29	27	26	25	24	23
B. Bound Tests											
B1. Real GDP growth	41	47	51	52	51	50	49	49	49	49	50
B2. Primary balance	41	45	45	44	42	39	36	34	33	31	30
B3. Exports	41	46	48	46	45	42	39	37	36	34	33
B4. Other flows 2/	41	46	47	45	44	41	39	37	35	33	32
B6. One-time 30 percent nominal depreciation	41	48	45	42	39	35	31	29	26	23	22
B6. Combination of B1-B5	41	44	44	42	40	37	35	33	31	30	29
C. Tailored Tests											
C1. Combined contingent liabilities	41	**57**	**56**	54	51	48	45	43	40	39	38
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**
PV of Debt-to-Revenue Ratio											
Baseline	**282.0**	**296.8**	**288.8**	**272.6**	**254.7**	**219.1**	**196.1**	**186.3**	**177.7**	**170.7**	**169.4**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	282	267	248	226	202	169	150	142	137	132	130
B. Bound Tests											
B1. Real GDP growth	282	316	344	336	322	288	269	268	269	271	281
B2. Primary balance	282	306	304	285	263	224	199	188	179	171	169
B3. Exports	282	310	320	302	281	241	215	205	195	187	185
B4. Other flows 2/	282	309	313	295	275	236	211	201	191	183	181
B6. One-time 30 percent nominal depreciation	282	325	303	275	245	201	171	156	142	129	123
B6. Combination of B1-B5	282	296	295	273	252	215	192	182	173	166	164
C. Tailored Tests											
C1. Combined contingent liabilities	282	388	373	351	324	277	245	233	223	214	212
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	**55.7**	**55.9**	**51.4**	**40.5**	**39.3**	**36.5**	**30.3**	**33.9**	**29.8**	**30.4**	**24.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	56	52	43	34	31	28	20	24	21	21	17
B. Bound Tests											
B1. Real GDP growth	56	59	60	49	49	46	40	47	44	46	40
B2. Primary balance	56	56	53	43	41	38	33	37	32	32	25
B3. Exports	56	56	52	41	40	37	31	34	31	32	25
B4. Other flows 2/	56	56	52	41	40	37	31	34	30	31	25
B6. One-time 30 percent nominal depreciation	56	54	52	42	41	38	32	34	31	31	25
B6. Combination of B1-B5	56	55	52	41	40	37	31	35	31	32	25
C. Tailored Tests											
C1. Combined contingent liabilities	56	56	70	48	45	41	48	41	34	34	28
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
2/ Includes official and private transfers and FDI.

Figure 1. Uganda: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, FY2020/21-2030/31 1/



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.2%	1.5%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	19	38
Avg. grace period	5	6

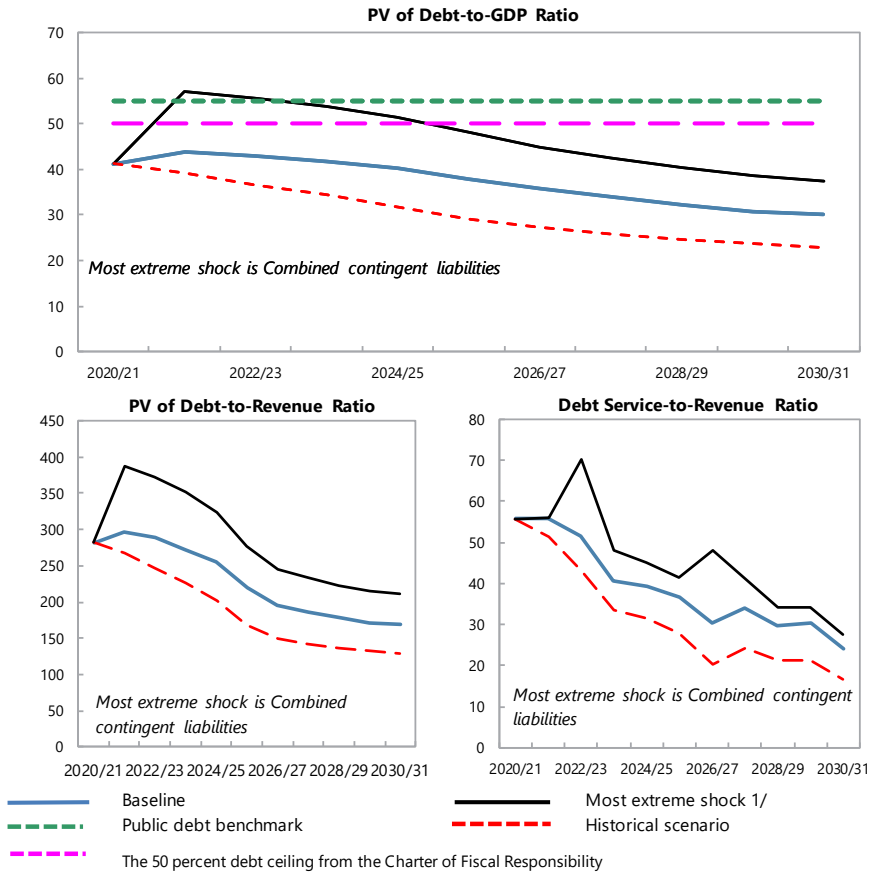
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Uganda: Indicators of Public Debt Under Alternative Scenarios, FY2020/21-2030/31



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	60%	60%
Domestic medium and long-term	27%	27%
Domestic short-term	13%	13%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.2%	2.2%
Avg. maturity (incl. grace period)	19	19
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	9.5%	9.5%
Avg. maturity (incl. grace period)	5	5
Avg. grace period	4	4
Domestic short-term debt		
Avg. real interest rate	6.0%	6.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

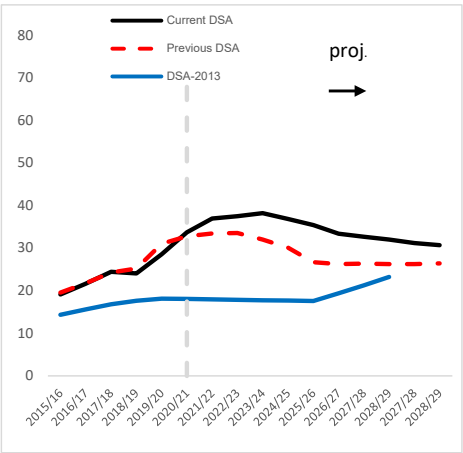
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

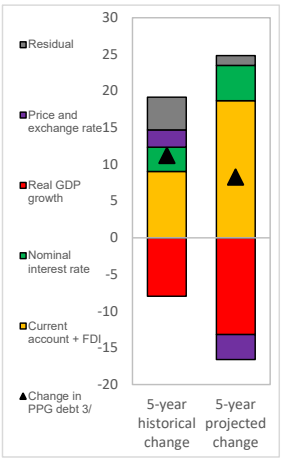
Figure 3. Uganda: Drivers of Debt Dynamics – Baseline Scenario

External debt

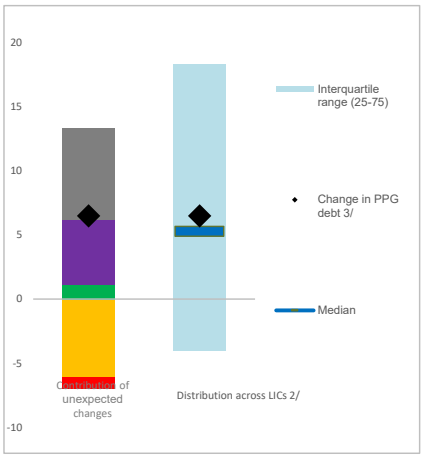
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

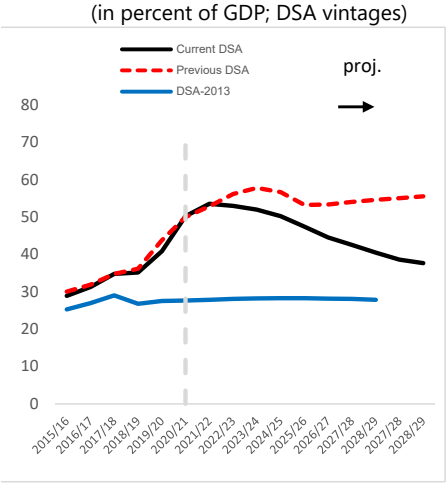


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

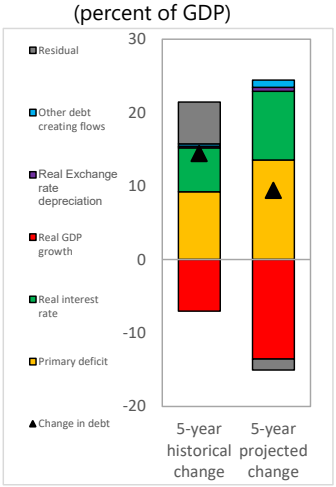


Public debt

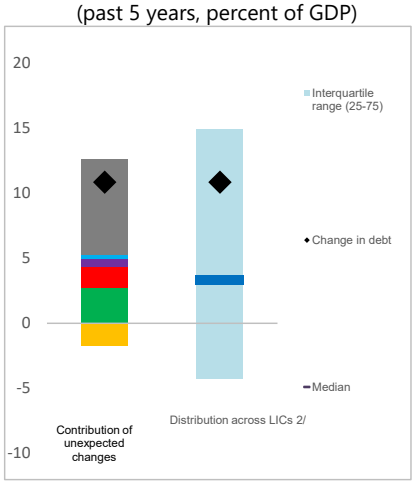
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



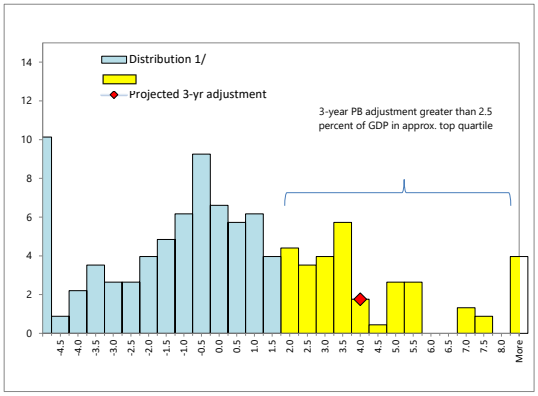
Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Difference between anticipated and actual contributions on debt ratios.
2/ Distribution across LICs for which LIC DSAs were produced.
3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

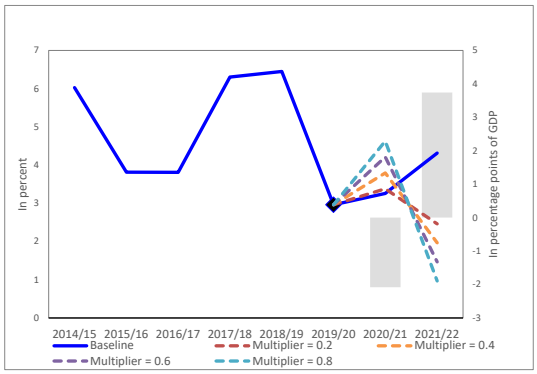
Figure 4. Uganda: Realism Tools

3-Year Adjustment in Primary Balance(Percentage points of GDP)



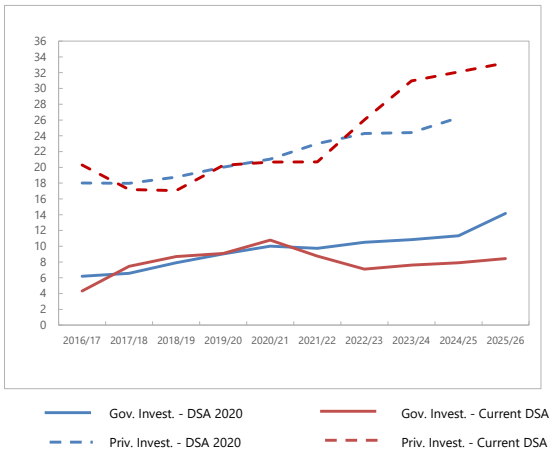
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

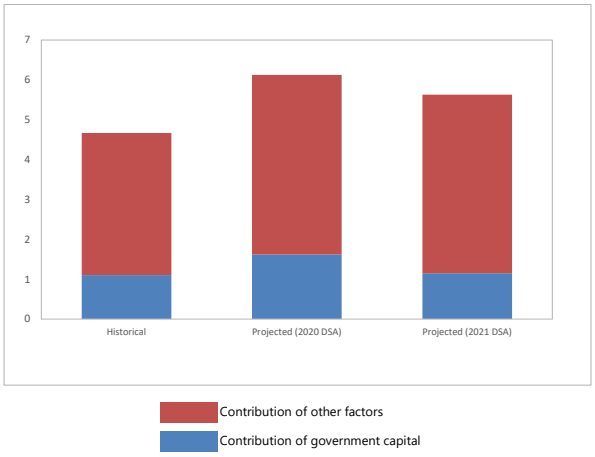


Figure 5. Uganda: Qualification of the Moderate Category, 2021-2031 1/

Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.